SIMULATION SOFTWARE FOR HOST-PARASITE COPHYLOGENIES: USER MANUAL

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cophylo.out is the executable file of a simulation software which takes a host phylogeny as input and outputs a simulated parasite phylogeny and host-parasite associations. The software is implemented in C.

Basic example

```
./cophylo.out -l speciation_rate -m extinction_rate -t TMRCA
```

cophylo.out takes several arguments and options using switches. In its most basic usage, you need to specify the speciation (switch -l) and extinction rates (switch -m) of the parasites, and the time to the most recent common ancestor of parasite species (switch -t). For example, to simulate parasites with a speciation rate $\lambda = 0.2$ and an extinction rate $\mu = 0.1$, and a TMRCA 4 Mya, and random cospeciation probability and host-switch rate, one enters:

```
./cophylo.out -l 0.2 -m 0.1 -t 4
```

It is assumed that three input files are present in the working directory —the list of edges and the list of edge length of the host phylogeny "host_edge.txt" and "host_edgelength.txt", and the distribution of the number of hosts per parasite "distrib.txt".
The software produces five output files —the list of edges and edge length of the parasite phyogeny "edges_0.txt" and "edgelength_0.txt", the list of host parasite associations "hpassoc_0.txt", the number of internal nodes in the parasite phylogeny "nnode_0.txt", and the cospeciation probability and host-switch rate "params_0.txt".

Input files format

Host edge list

The host edge list is a space or tabulation-delimited 2 column-table of integer values, where each row represents an edge; the element on the first column represents the ancestral node and the element on the second column represents the child node. Leaves are numbered from 1 to n, the root is numbered $n + 1$, and internal nodes are numbered from $n + 2$ to $2n - 1$. The first edge has to have the root as first element.

For the previous example, a possible input file "host_edge.txt" would be:

```
4 5
5 1
5 2
4 3
```

This input file corresponds to a phylogeny with 3 leaves, where species 1 and 2 share a more recent common ancestor (node 5) than species 1 and 3, or 2 and 3. The root is node 4, and the first edge (4 5) has the root as first element.

Host edge length list

The host edge length list is a single column table of positive values; the elements represent the length of each edge from file "host_edge.txt". There are $2(n - 1)$ values, where n is the number of host species in the phylogeny.

For the previous example, a possible input file "host_edgelength.txt" would be:

```
3.00
2.00
2.00
5.00
```

Distribution of number of hosts per parasite

The distribution of number of hosts per parasite is a space or tab-delimited list of values. Element i represent probability of a parasite infecting i hosts, so each element is in $[0, 1]$ and elements of the list sum to 1.

For the previous example, a possible input file "distrib.txt" would be:

```
0.8 0.1 0.05 0.05
```

This input file corresponds to a parasite phylogeny where species have a probability 0.8 to be specialists (infect a single host species), and a probability 0.2 to be generalists (infect several host species), with parasite species infecting 2 host species with probability 0.1, infecting 3 host species with probability 0.05, and 4 host species with probability 0.05.

Output files

Parasite edge list

The parasite edge list has the same format as the input host edge list. One output parasite edge list file is written for each simulation; each file is named "edges_i.txt", with i ranging from 0 to the number of simulations minus 1.

For the previous example, a possible output file "edges_0.txt" would be:

```
5 6
5 7
6 1
6 2
7 3
7 4
```

Parasite edge length list

The parasite edge length follows the same format as the input host edge length list file. One output parasite edge length list file is written for each simulation; each file is named "edgelength_i.txt", with i ranging from 0 to the number of simulations minus 1.

For the previous example, a possible output file "edgelength_0.txt" would be:

```
2.50
2.50
2.00
2.00
2.00
2.00
```

Host-parasite associations list

The host-parasite associations list is a tab-delimited 2 column table, where each row corresponds to the association between one host and one parasite. Elements in the first column represent the parasite ID, numbered from $2n$ to $2n + n_p$, where n is the number of host species, and n_p the number of parasite species; elements in the second column represent the host ID, numbered from 1 to n. This numbering ensures that hosts and parasites do not have any identical ID. One output host-parasite associations file is written for each simulation; each file is named "hpassoc_i.txt", with i ranging from 0 to the number of simulations minus 1.

For the previous example, a possible output file "hpassoc_0.txt" would be:

```
6 1
7 2
8 1
9 2
```

This output file corresponds to an association between parasite 6 and host 1, between parasite 7 and host 2, between parasite 8 and host 1, and between parasite 9 and host 2.

Number of internal nodes in parasite phylogeny

The number of internal nodes file contains a integer single number. One output number of internal nodes file is written for each simulation; each file is named "nnode_i.txt", with i ranging from 0 to the number of simulations minus 1.

Parameters

The parameter file is a tab-delimited file with two values; the first value corresponds to the cospeciation probability, and the second value corresponds to the host-switch rate. One output parameter file is written for each simulation; each file is named "params_i.txt", with i ranging from 0 to the number of simulations minus 1.

For the previous example, a possible output file "params_0.txt" would be:

```
0.3678 0.5894
```

This output file corresponds to a simulation with a cospeciation probability of 0.3678, and a host-switch rate of 0.5894.

Command line options

Table 1: **Summary of switches**

Switch	Model parameter	Default	Description
-l	λ	0.96	parasite speciation rate
-m	μ	0.59	parasite extinction rate
-c	p_{co}	Unif([0,1])	cospeciation probability
-s	ν	Unif([0,1])	host-switch rate
-t	-	13.97	parasite TMRCA
-n	-	1	number of simulations
-N	-	50	minimum number of parasite species
-P	-	"distrib.txt"	input file for distribution of number of hosts per parasite
-i	-	""	suffix for input files
-o	-	""	suffic for output files
-S	-	current time	seed for random number generator

A more complex example: the bird-malaria cophylogeny

In this example, we want to simulate the avian malaria parasite phylogeny and associations between malaria parasite species and bird species, with random cospeciation probability and host-switch rate. We will simulate a single phylogeny (option -n 1, or no switch as 1 simulation is the default). The parasite speciation rate is assumed to be 0.96 (option -l 0.96), and the extinction rate 0.59 (option -m 0.59). The TMRCA of parasites is assumed to be 13.96 Mya (option -t 13.96), The input files for the parasite bird phylogeny are "examples/input/bird-malaria_host_edgelength.txt" and "examples/input/bird-malaria_host_edge.txt" (switch -i examples/input/bird-malaria_). The output files will be written in "examples/output/bird-malaria_edgelength_0", "examples/output/bird-malaria_edges_0", "examples/output/bird-malaria_hpassoc_0", "examples/output/bird-malaria_nnode_0", and "examples/output/bird-malaria_params_0" (option -o examples/output/bird-malaria_). We constrain simulation to contain at least 50 parasite species (option -N 50; simulation will be rerun until fulfilling this condition). We take a seed of 3 (option -S 3). The input distribution of the number of hosts per parasite is in file "examples/input/bird-malaria_distribHperP.txt" (option -P examples/input/bird-malaria_distribHperP.txt)

The full command is the following:

```
./cophylo.out -l 0.96 -m 0.59 -t 13.96 -i examples/input/bird-malaria_
              -o examples/output/bird-malaria_ -N 50 -S 3
              -P examples/input/bird-malaria_distribHperP.txt
```